

NYC Taxi Trip Duration — Regression Report

23CSE301 Machine Learning | Review 1 | Regression track

Data file: /Users/bettim/Documents/Semester 5/Machine Learning/Capstone Evaluation/data/nyc_taxi_trip_duration.csv

Raw rows loaded: 352,000

Rows after cleaning: 349,080

Rows in modelling table: 349,080

Training rows: 279,264 Evaluation rows: 50,000

FAST_DEV = False

Best model by test R^2 : Gradient Boosting Regressor

Official models trained: 12

Figures exported: 25

Pipeline: audit → EDA → clean → engineer → split/encode/scale

→ 10 models → compare/tune/CV → manual tester.

Encoders and scalers were fit on the training split only.

dropoff_datetime and id were never used as predictors.

Data cleaning — rows removed at each step

step	rows_before	rows_after	rows_removed	pct_removed	note
loaded	352000	352000	0	0.0000	raw file
drop duplicate rows	352000	352000	0	0.0000	exact row duplicates
drop leakage / id columns	352000	352000	0	0.0000	id + dropoff_datetime removed (column dropped)
NYC bounding box	352000	351601	399	0.1134	$[-74.30,-73.70] \times [40.50,40.90]$
passenger_count 1–6	351601	351589	12	0.0034	drop 0 and 7+
duration 60–7200 s	351589	349080	2509	0.7136	drop cancelled / multi-day trips

Test metrics (same held-out trips for every model)

Rank	Model	Train rows	R2	RMSE	MAE	CV R2 mean	CV R2 std	Fit seconds
1	Gradient Boosting Regressor	279264	0.7961	291.4826	188.7737	0.7921	0.0028	2.4531
2	Random Forest Regressor	279264	0.7921	294.3231	189.9446	0.7860	0.0022	18.2844
3	Random Forest Regressor (tuned)	279264	0.7912	294.9744	189.7353	0.7846	0.0025	21.1713
4	Decision Tree Regressor	279264	0.7506	322.3702	211.5239	0.7451	0.0025	2.2113
5	Support Vector Regressor	12000	0.7281	336.6159	213.1370	0.7204	0.0106	3.0475
6	Polynomial Regression	279264	0.7026	352.0505	238.5700	0.6958	0.0040	2.5130
7	K-Nearest Neighbors Regressor	40000	0.6993	354.0092	243.5402	0.6854	0.0042	0.0008
8	Ridge Regression	279264	0.6241	395.7908	273.0075	0.6208	0.0039	0.0363
9	Ridge Regression (tuned)	279264	0.6241	395.7913	273.0072	0.6208	0.0039	0.0136
10	Linear Regression	279264	0.6241	395.7913	273.0072	0.6208	0.0039	0.0723
11	Lasso Regression	279264	0.6240	395.8115	273.1529	0.6206	0.0040	2.4220
12	ElasticNet Regression	279264	0.6092	403.5481	281.5870	0.6047	0.0039	0.0999
13	Mean baseline	279264	-0.0000	645.5262	462.3763	-0.0000	0.0000	1.6420

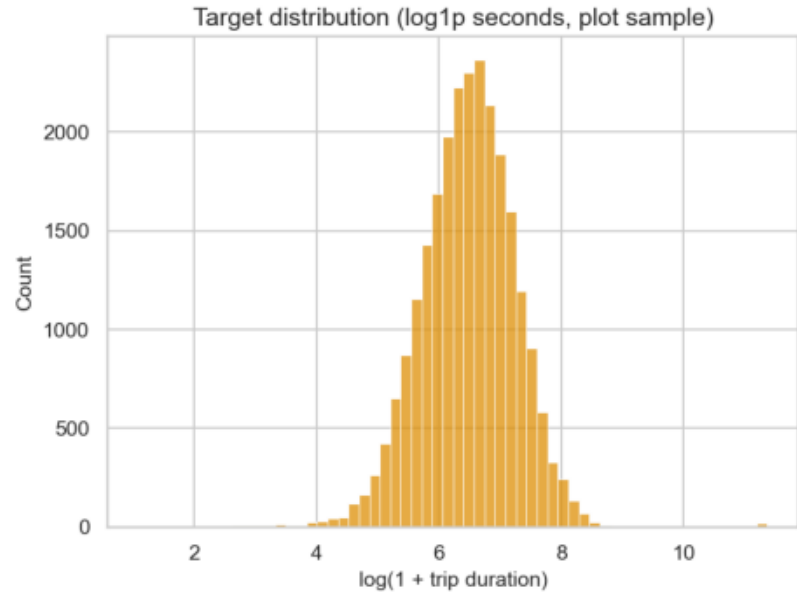
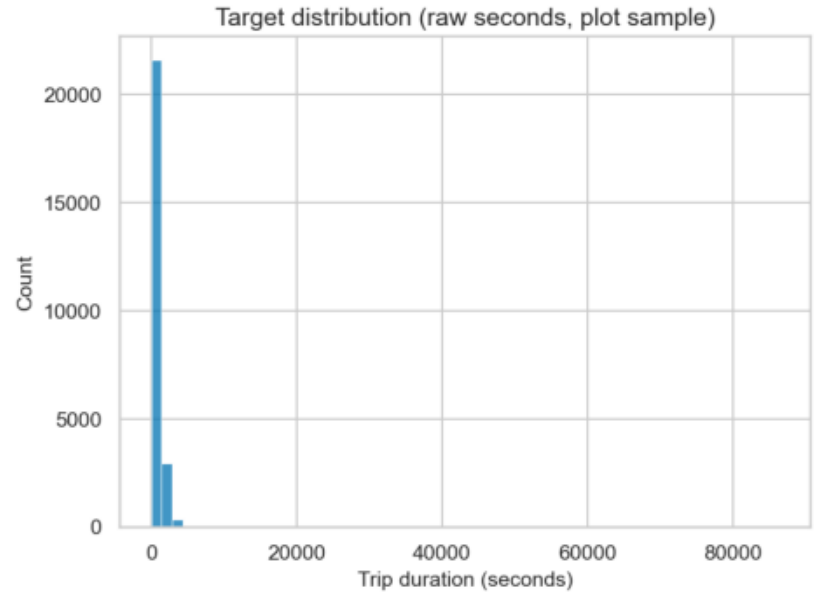
5-fold CV R² for the two best official models

Model	Test R2	Test RMSE	Test MAE	CV folds	CV R2 mean	CV R2 std	Fold scores
Gradient Boosting Regressor	0.7961084274141603	291.4825502294567	188.77367672215055	5	0.7921	0.0028	0.7948, 0.7867, 0.7938, 0.7927, 0.7945
Random Forest Regressor	0.7911940842113163	294.9743971246077	189.73529144208754	5	0.7846	0.0025	0.7868, 0.7798, 0.786, 0.7847, 0.786

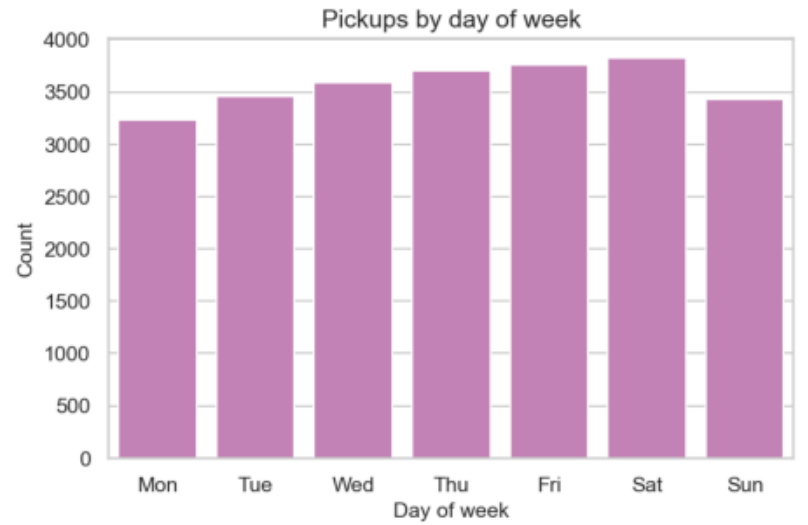
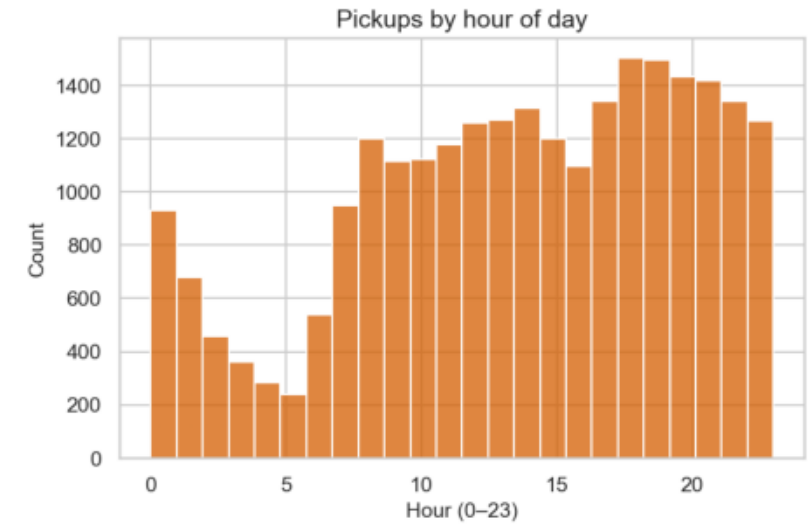
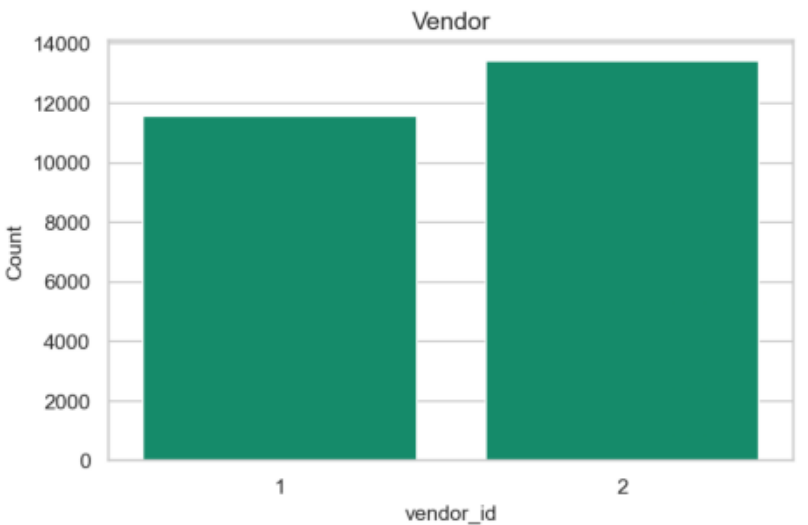
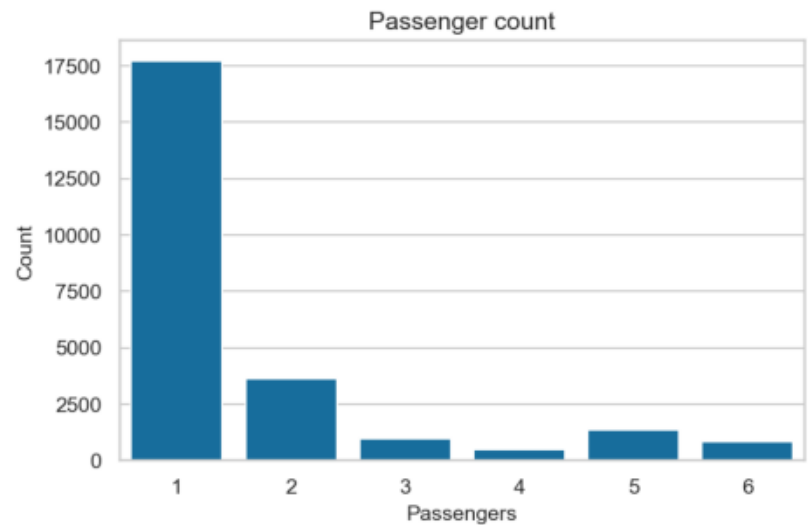
Manual tester — predicted duration for user-edited trips

trip	model_used	predicted_seconds	predicted_minutes	trip_distance_km	pickup_hour	is_rush_hour
1	Gradient Boosting Regress	675.8615	11.2644	1.5041	17	1
2	Gradient Boosting Regress	630.7121	10.5119	1.8075	0	0

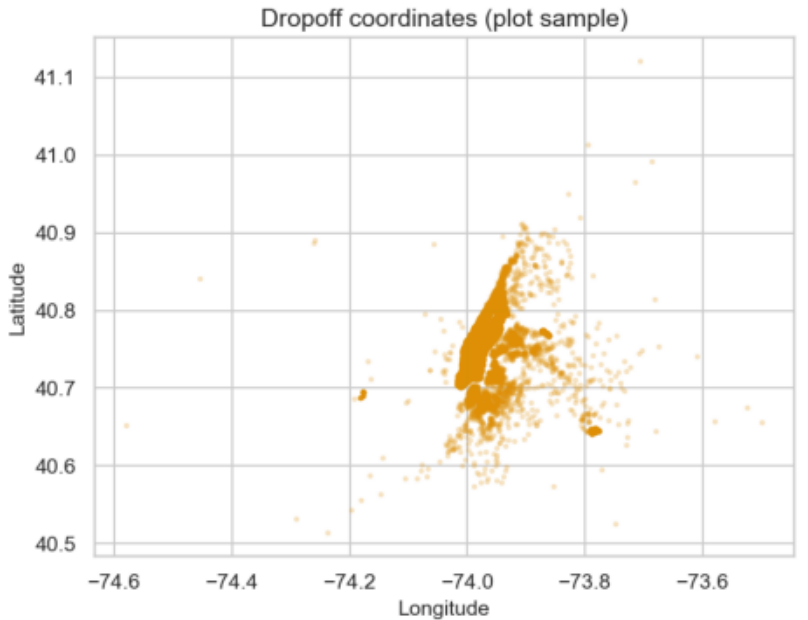
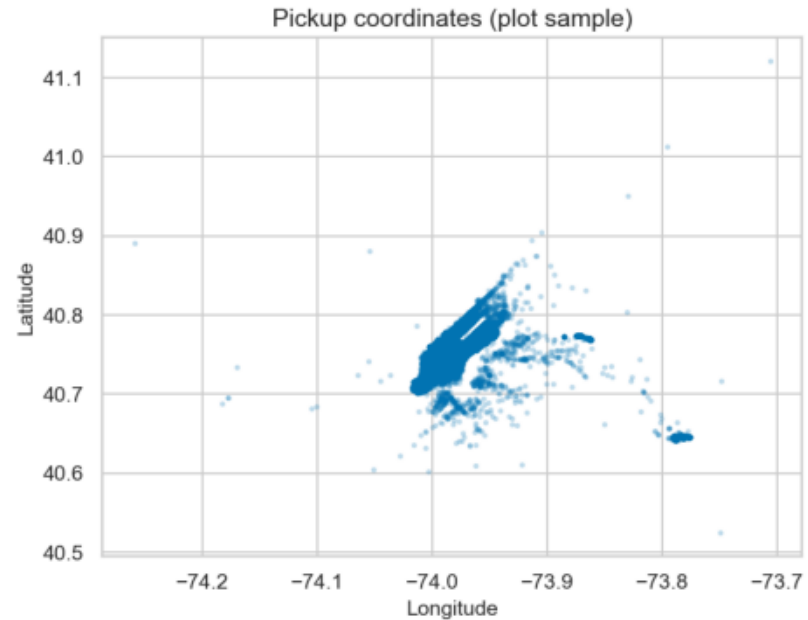
01 target distribution raw seconds plot sample



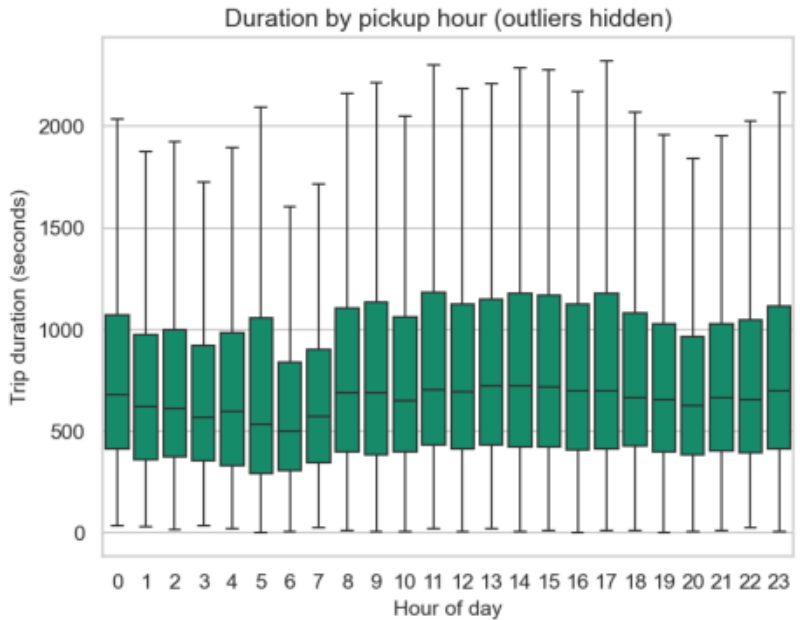
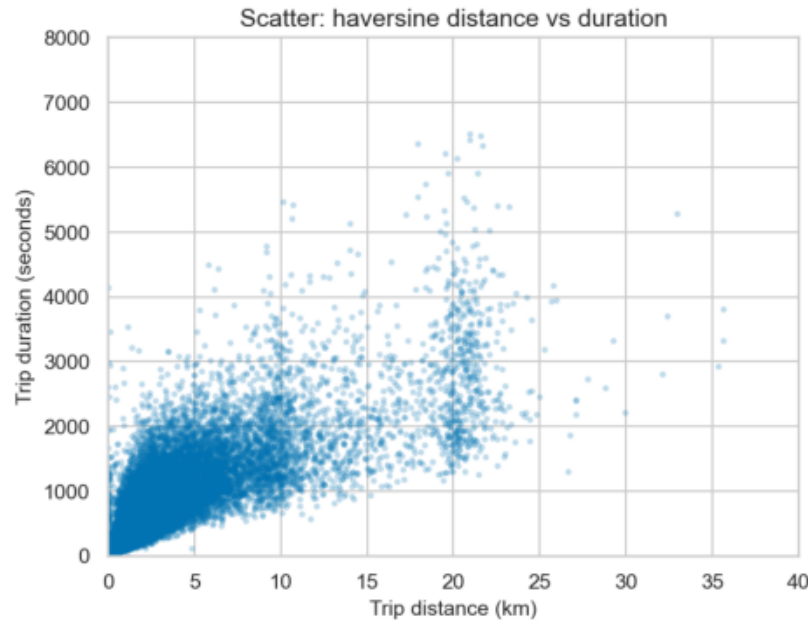
02 passenger count

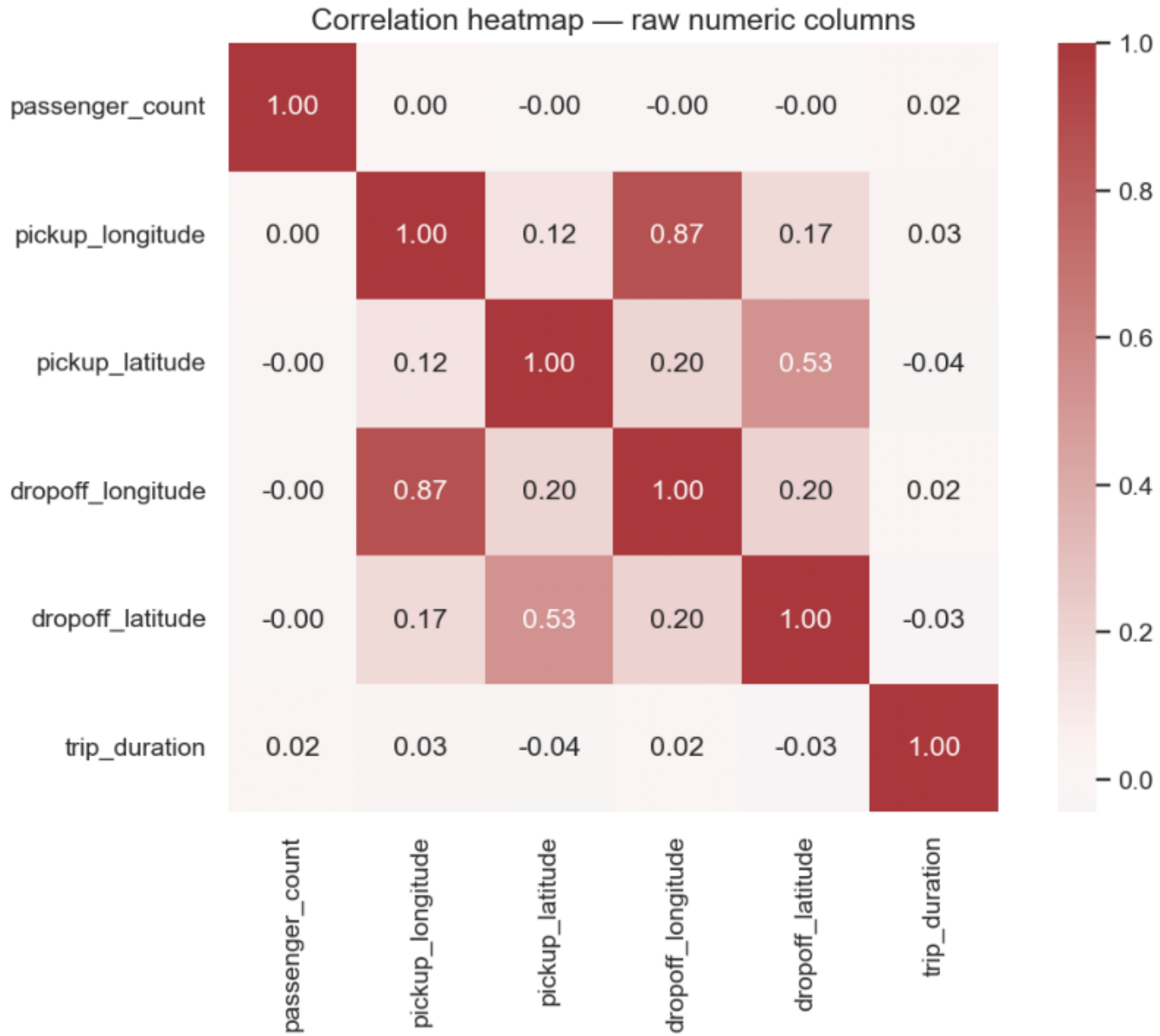


03 pickup coordinates plot sample

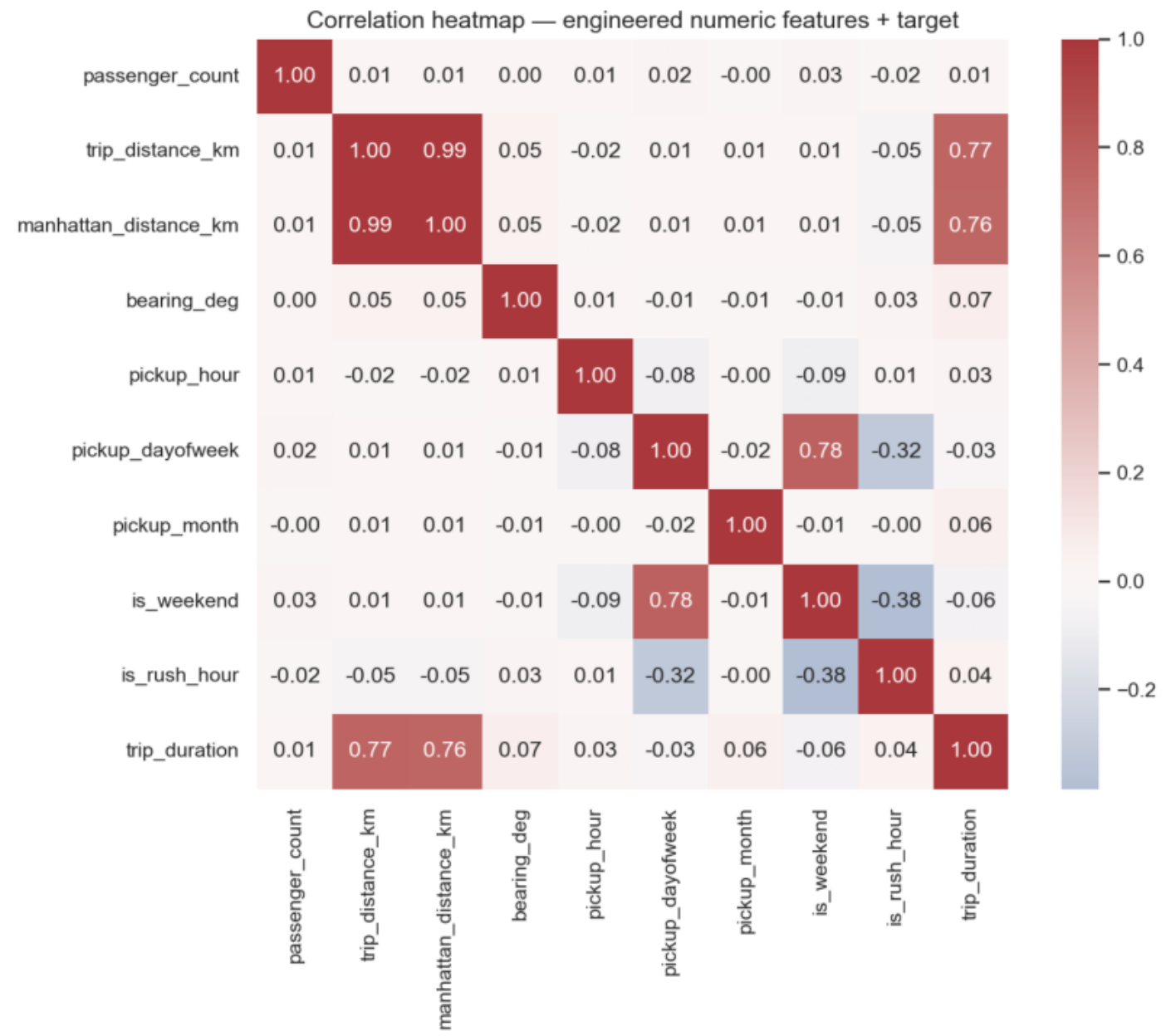


04 scatter haversine distance vs duration

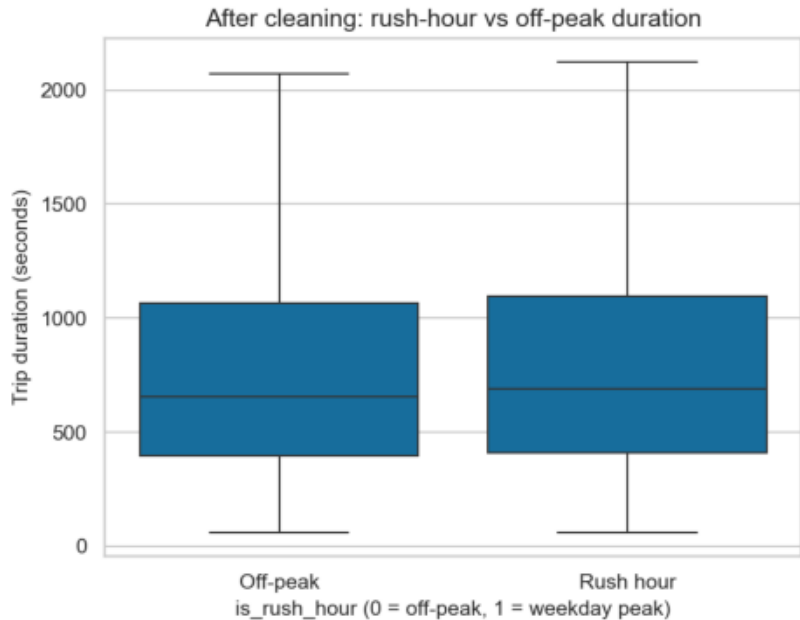
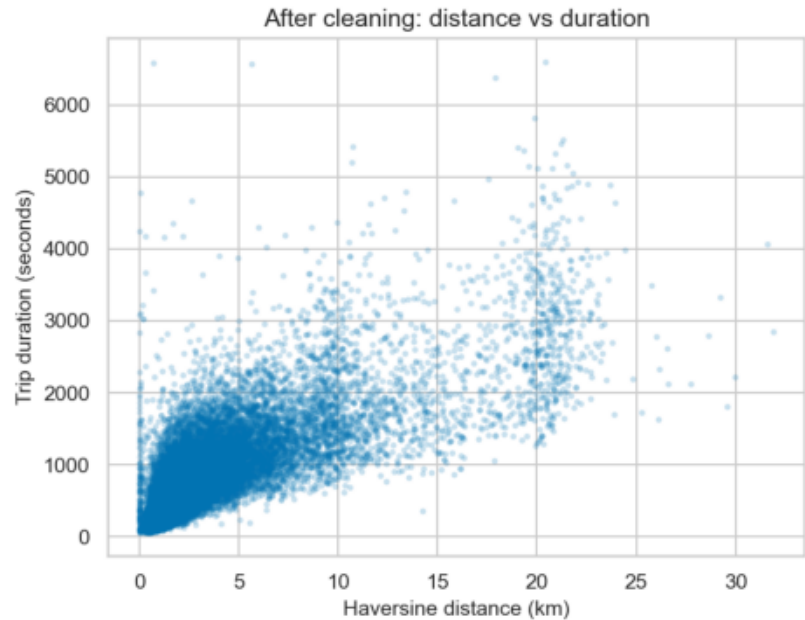




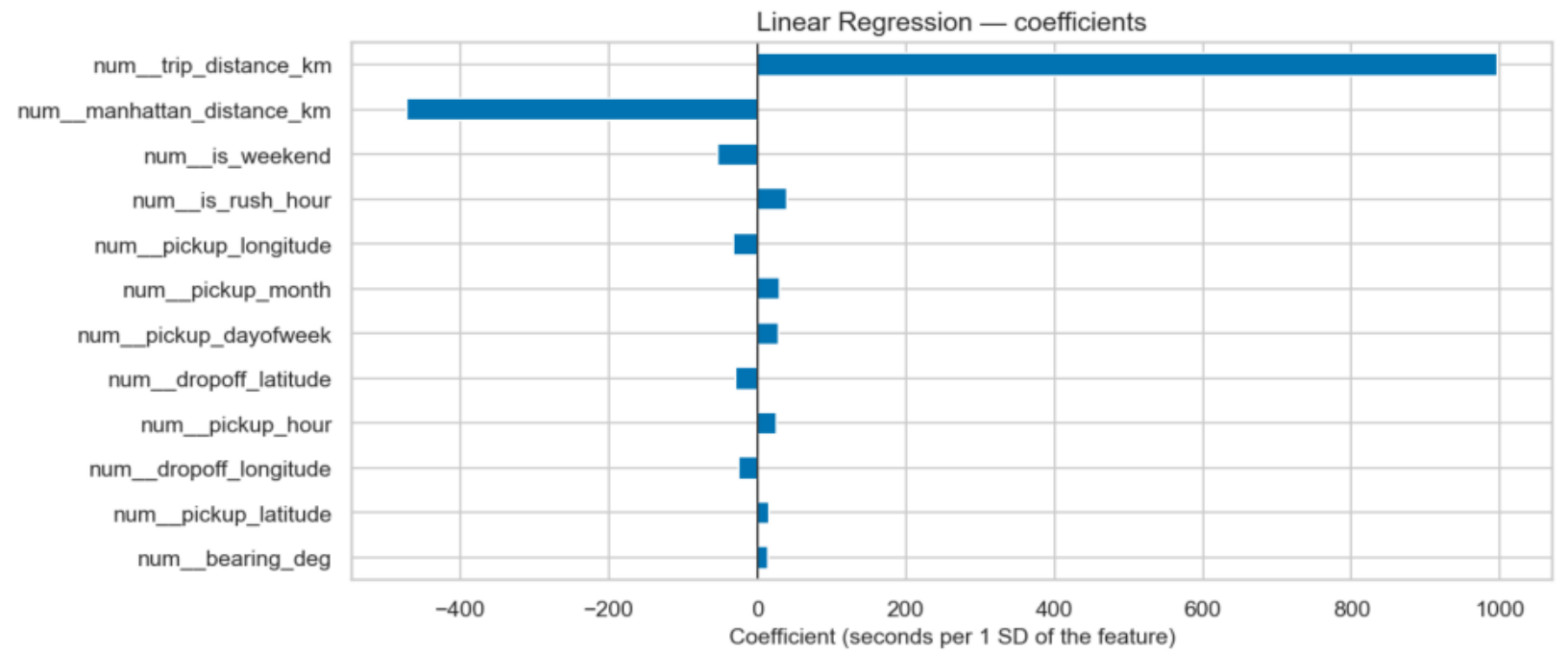
06 correlation heatmap engineered numeric features target



07 after cleaning distance vs duration



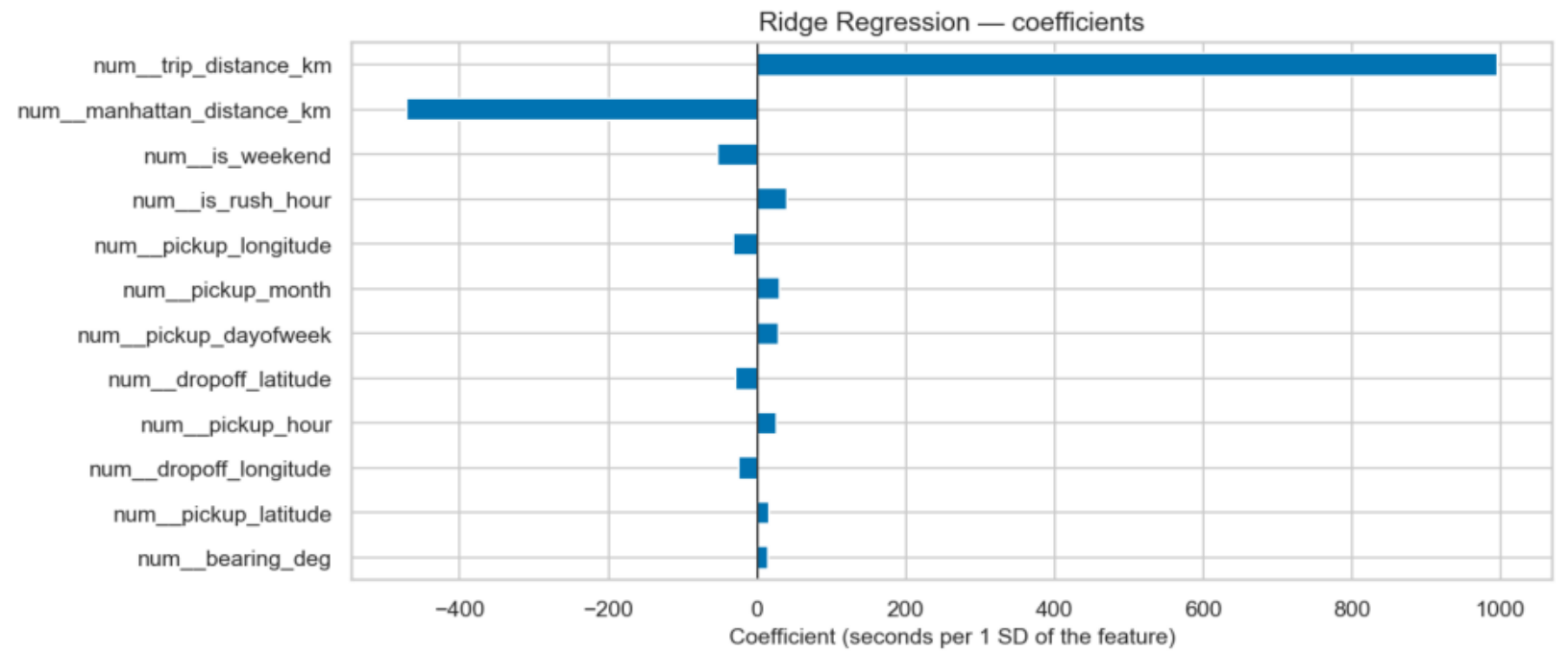
08 linear regression result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Linear Regression	0.6241	395.8	273.0	0.6208	0.0039

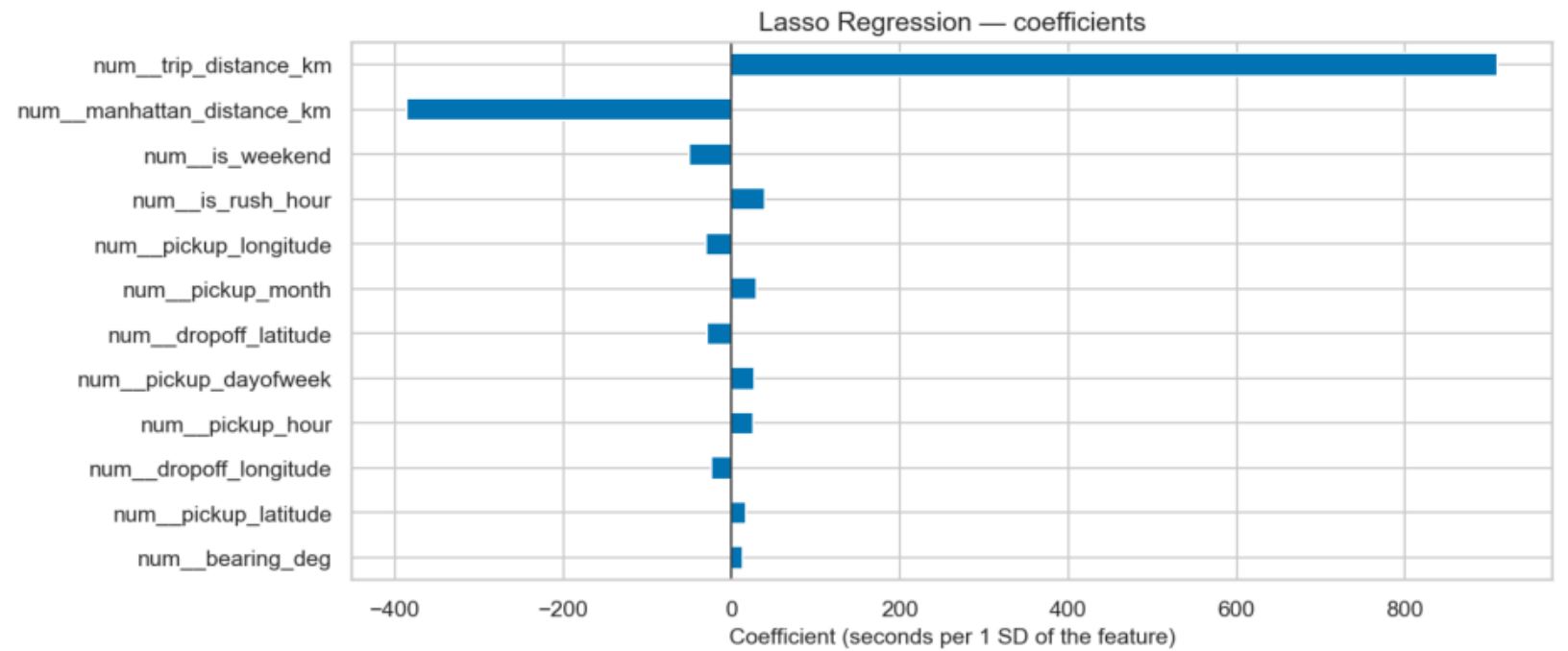
09 ridge regression result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Ridge Regression	0.6241	395.8	273.0	0.6208	0.0039

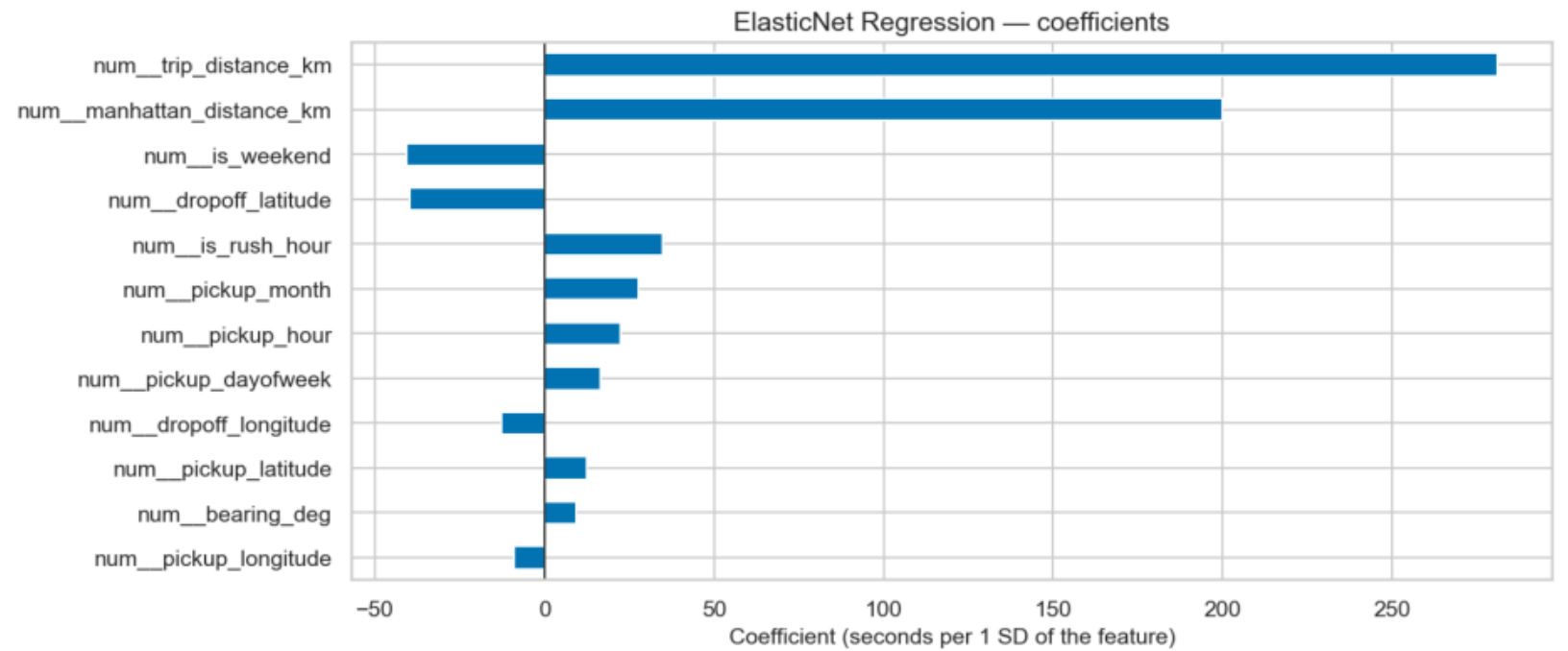
10 lasso regression result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Lasso Regression	0.6240	395.8	273.2	0.6206	0.0040

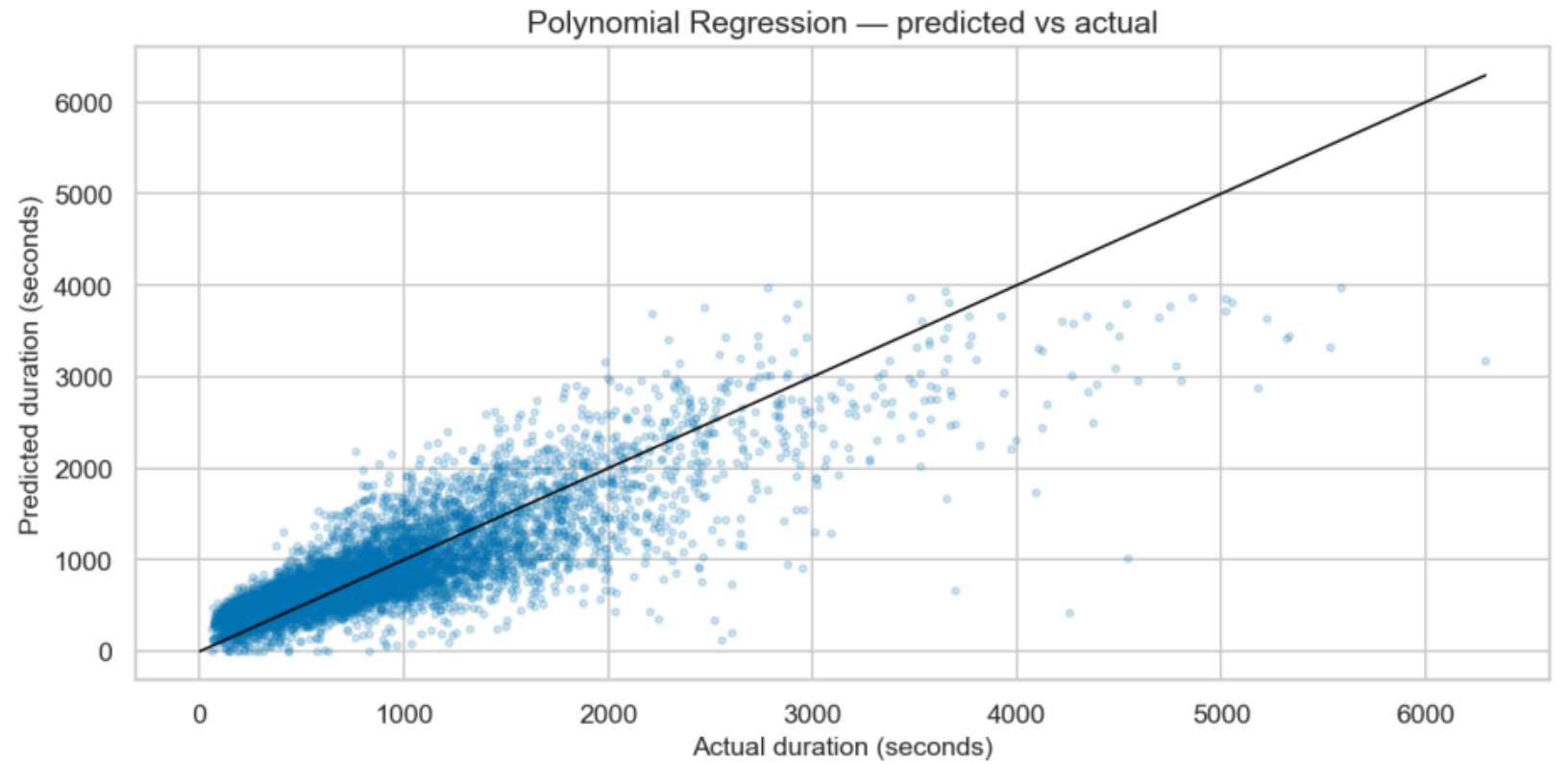
11 elasticnet regression result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
ElasticNet Regression	0.6092	403.5	281.6	0.6047	0.0039

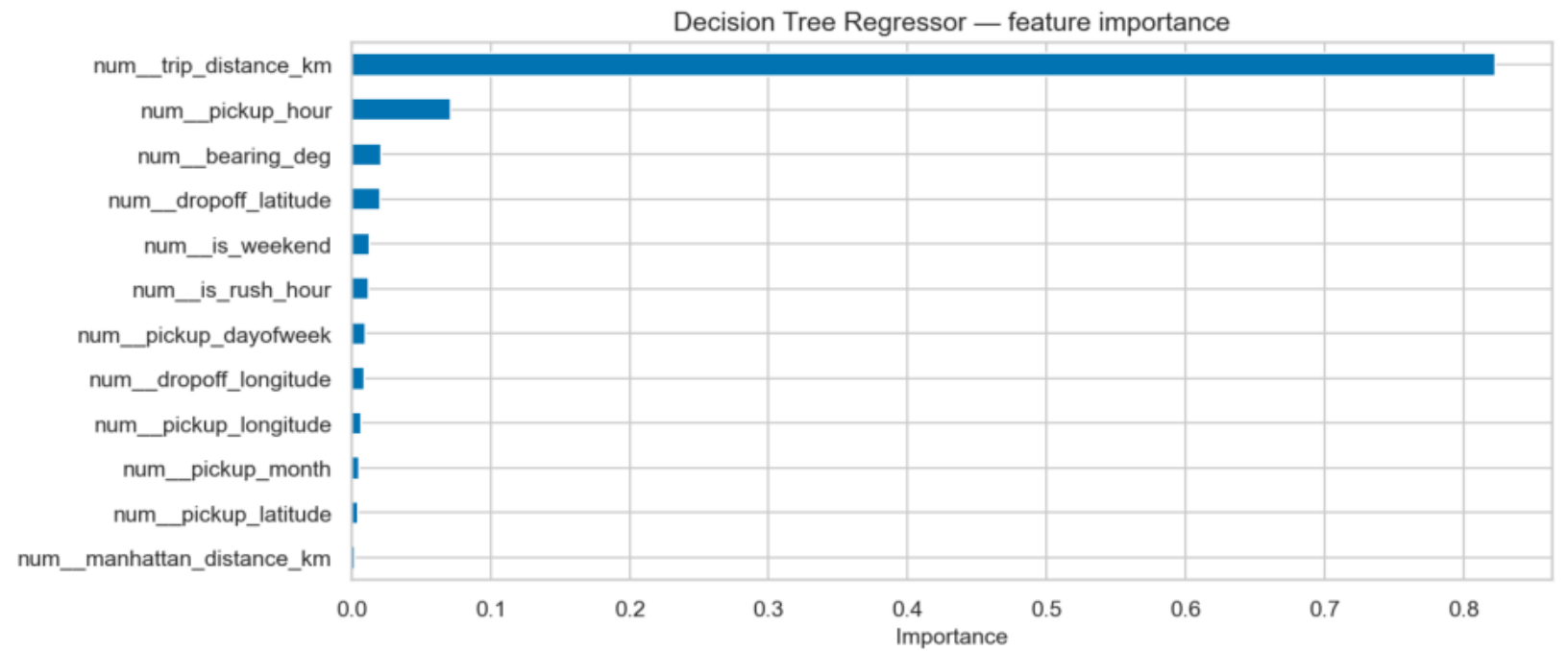
12 polynomial regression result



Mandatory metrics (shared test set)

Model	Test R ²	RMSE (s)	MAE (s)	5-fold CV R ² mean	5-fold CV R ² std
Polynomial Regression	0.7026	352.1	238.6	0.6958	0.0040

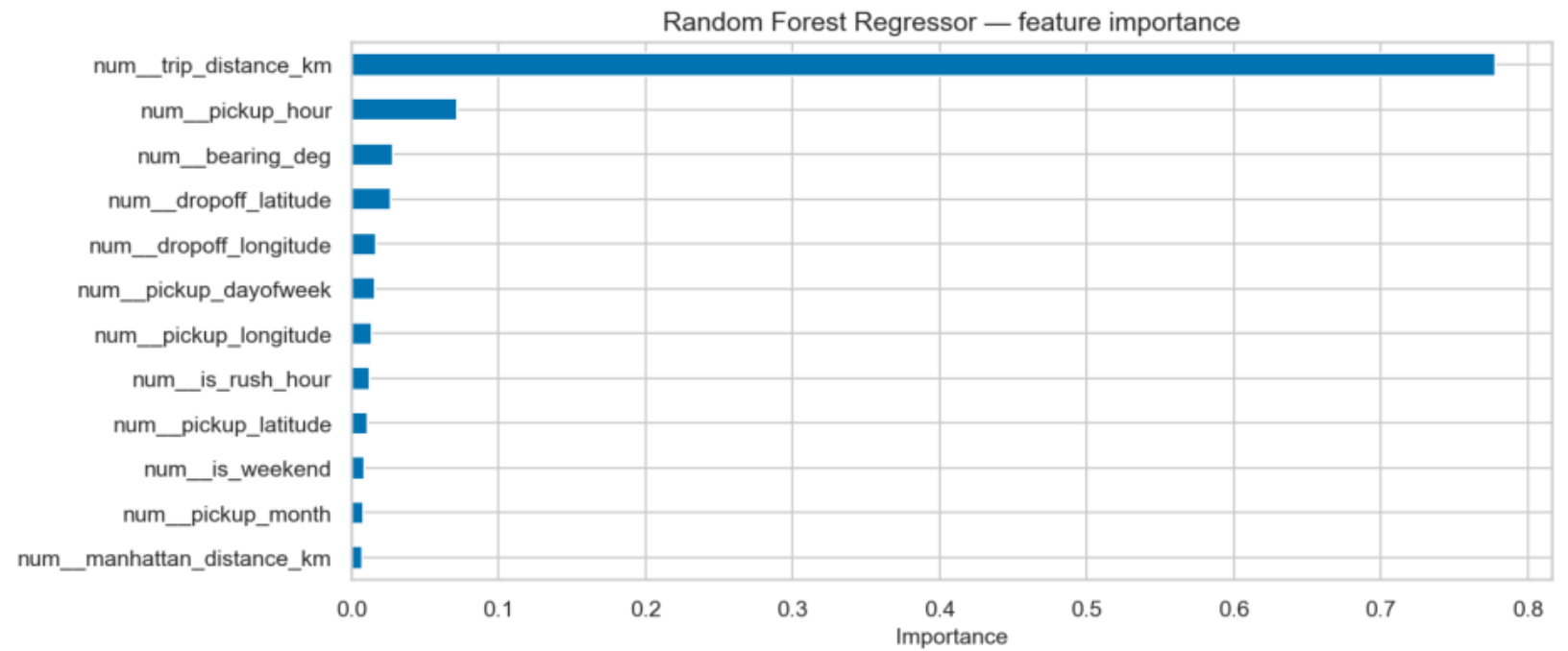
13 decision tree regressor result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Decision Tree Regressor	0.7506	322.4	211.5	0.7451	0.0025

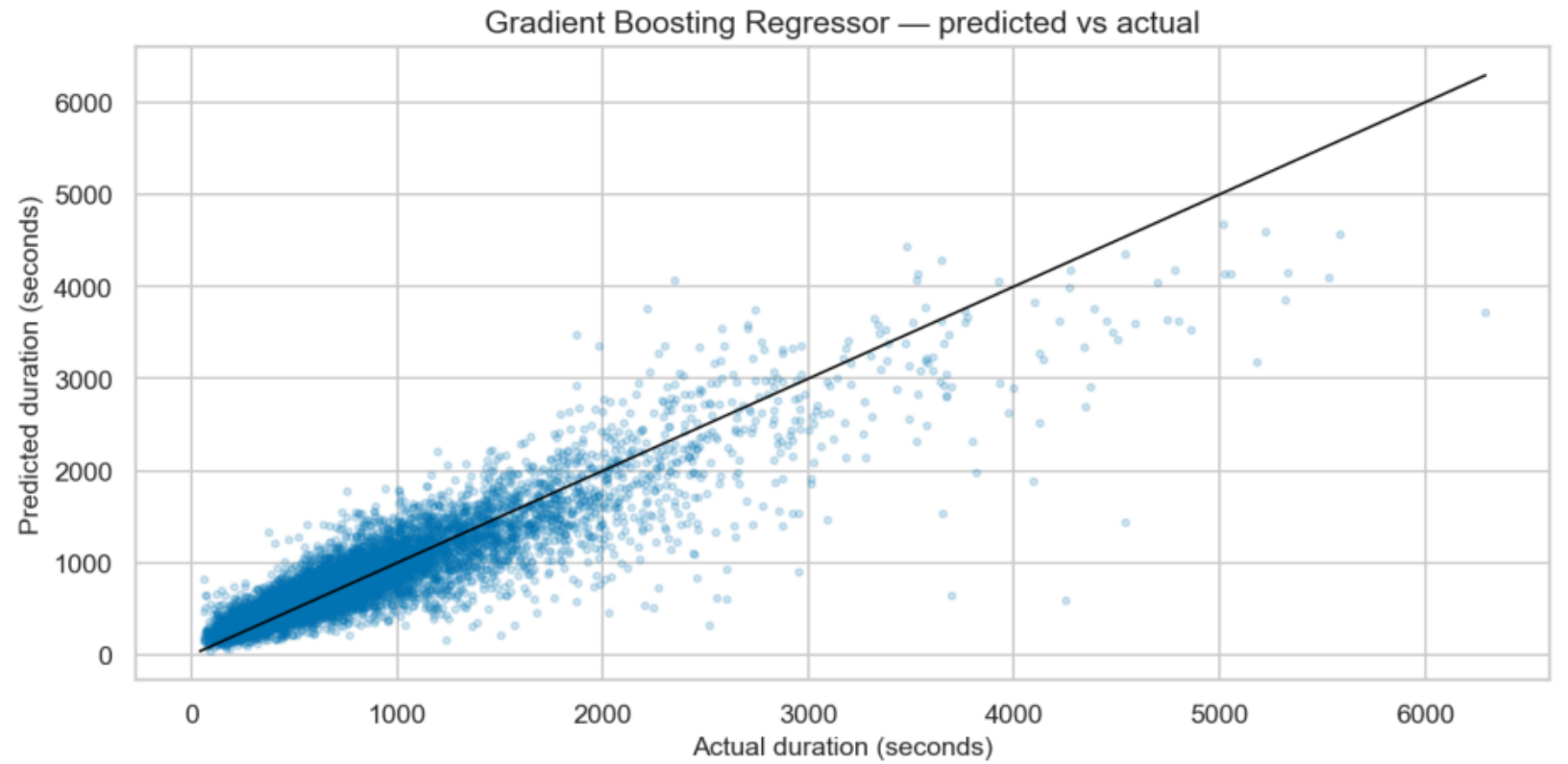
14 random forest regressor result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Random Forest Regressor	0.7912	295.0	189.7	0.7846	0.0025

15 gradient boosting regressor result



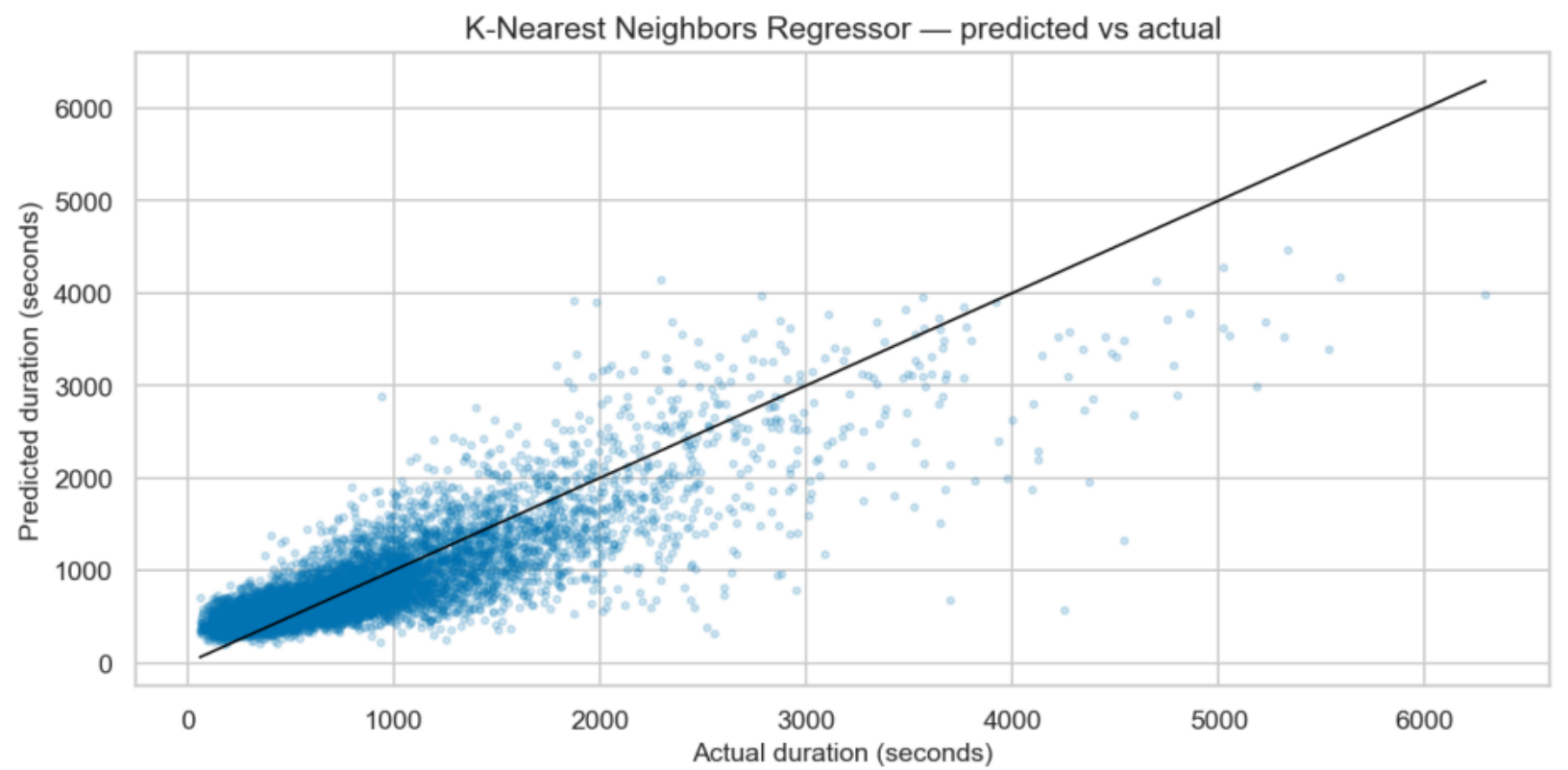
Mandatory metrics (shared test set)

Model	Test R ²	RMSE (s)	MAE (s)	5-fold CV R ² mean	5-fold CV R ² std
Gradient Boosting Regress	0.7961	291.5	188.8	0.7921	0.0028



Mandatory metrics (shared test set)

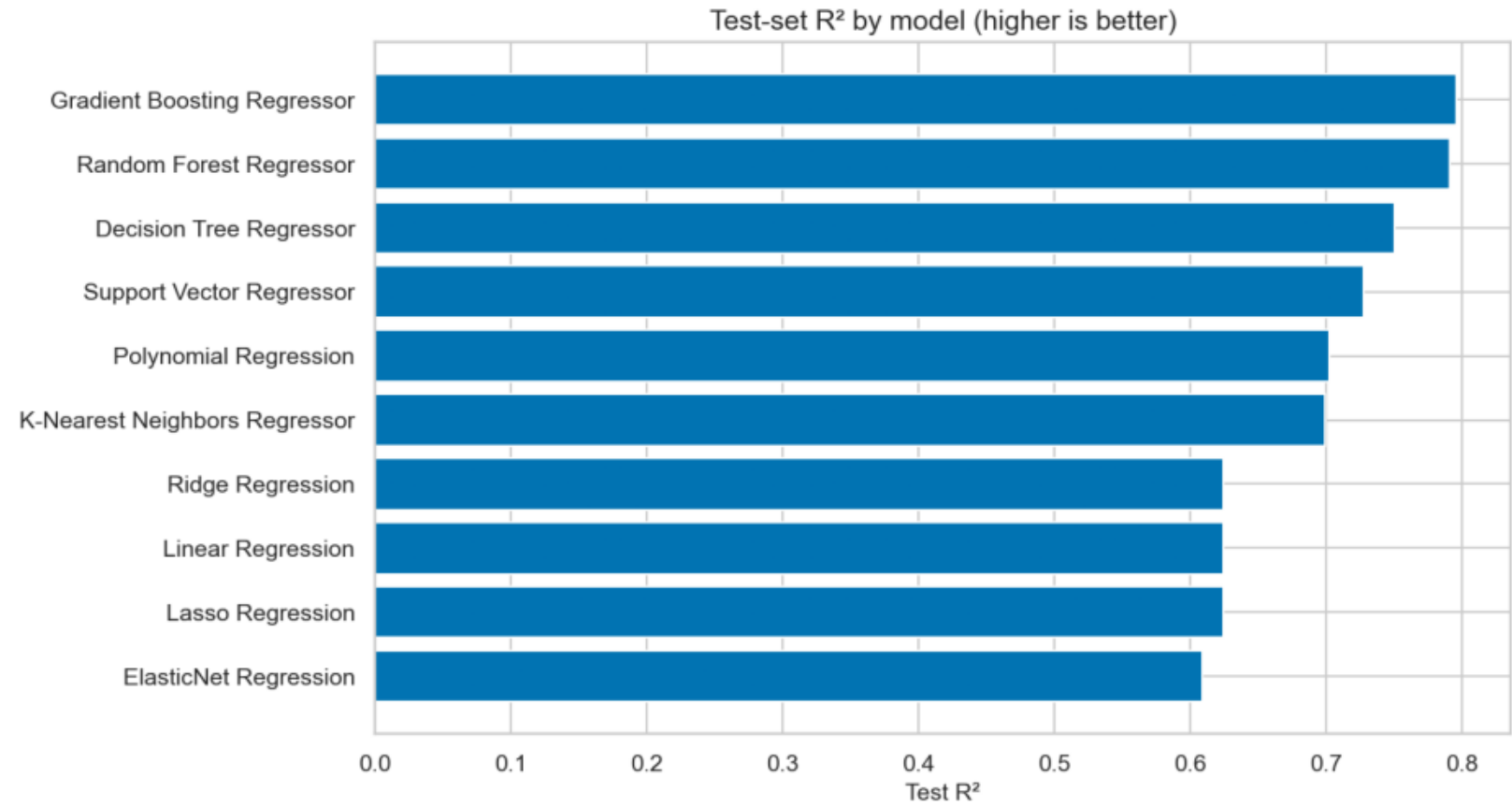
Model	Test R ²	RMSE (s)	MAE (s)	5-fold CV R ² mean	5-fold CV R ² std
Support Vector Regressor	0.7281	336.6	213.1	0.7204	0.0106



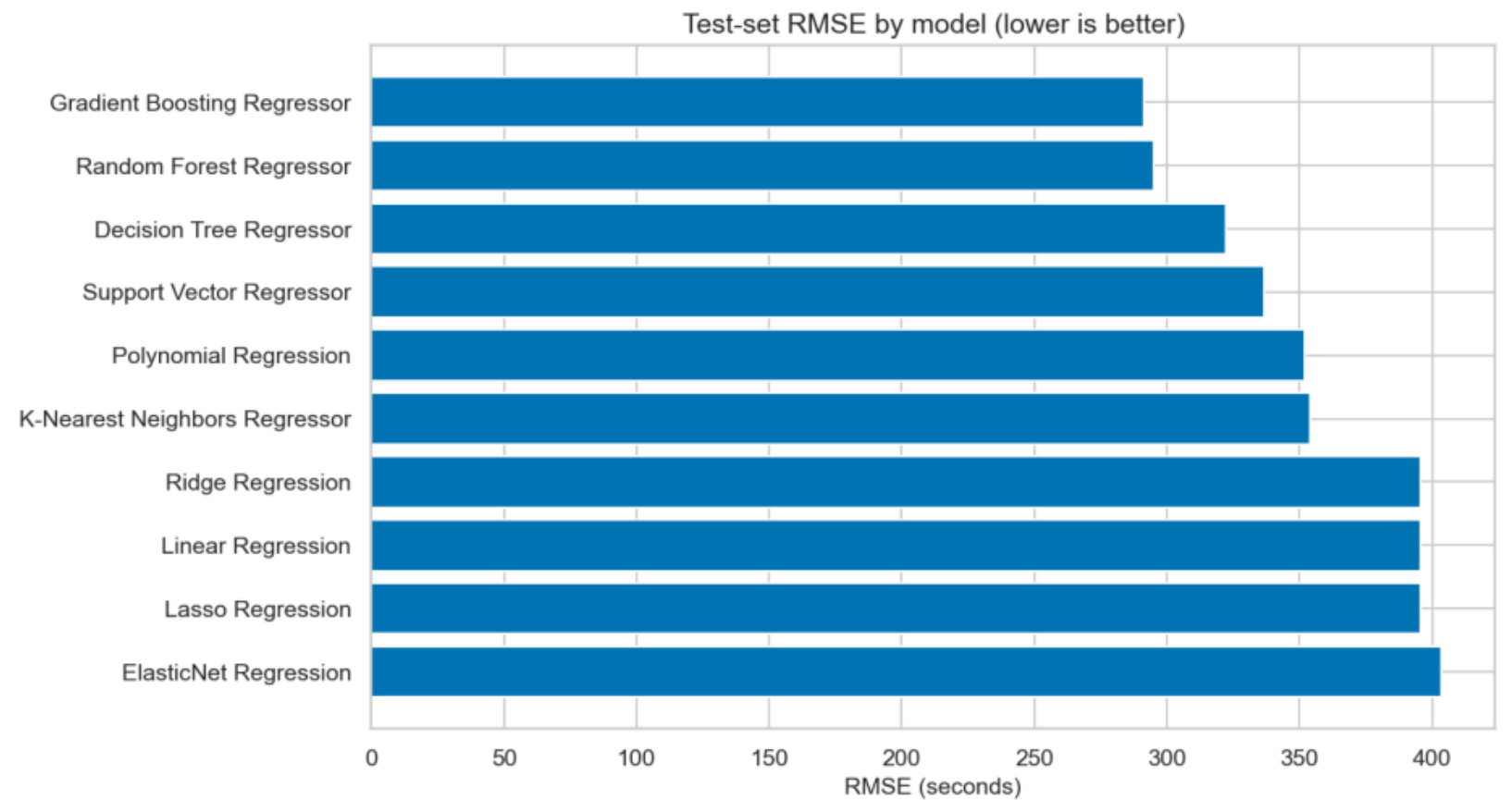
Mandatory metrics (shared test set)

Model	Test R ²	RMSE (s)	MAE (s)	5-fold CV R ² mean	5-fold CV R ² std
K-Nearest Neighbors Regres	0.6993	354.0	243.5	0.6854	0.0042

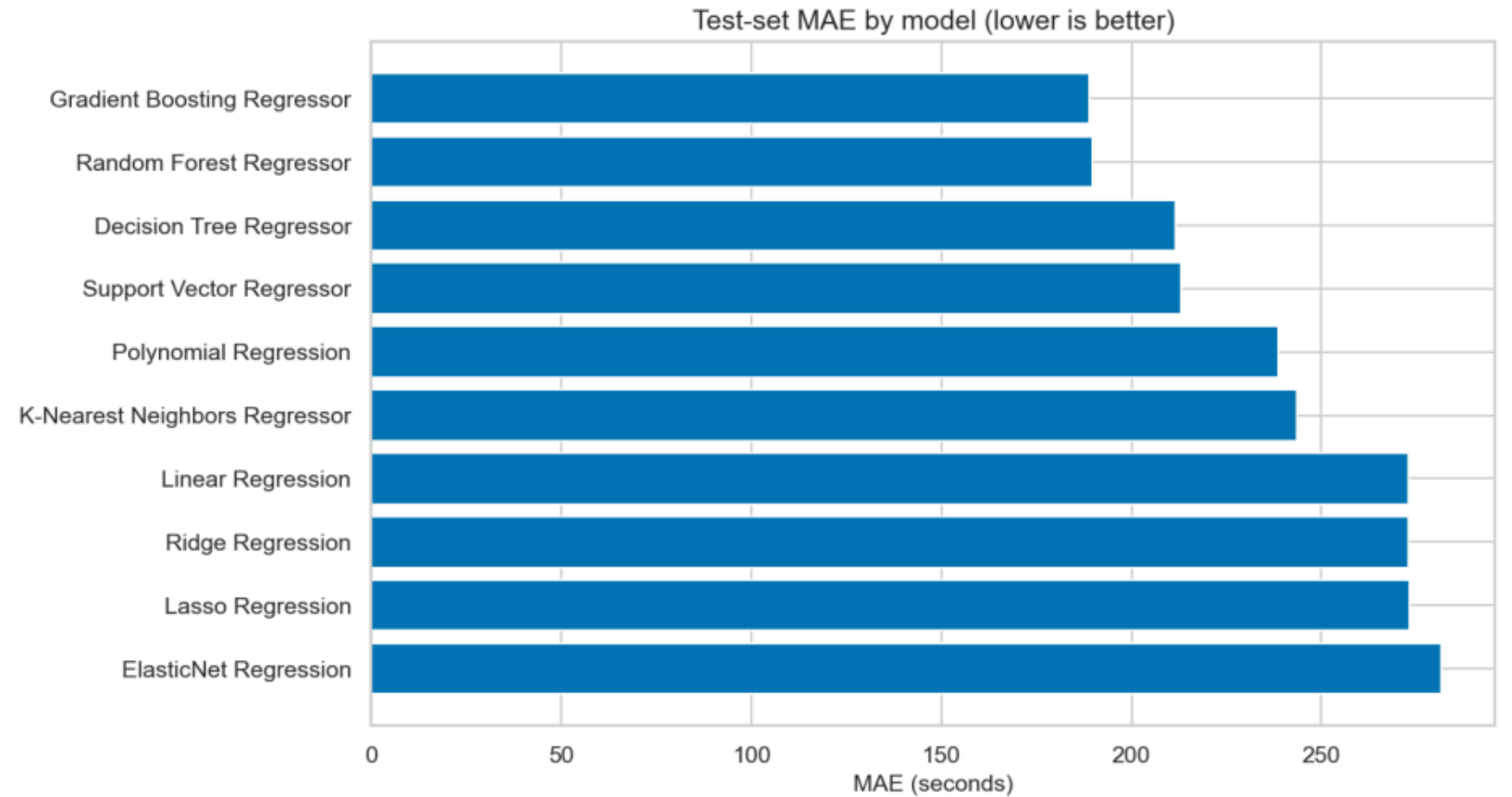
18 test set r^2 by model higher is better



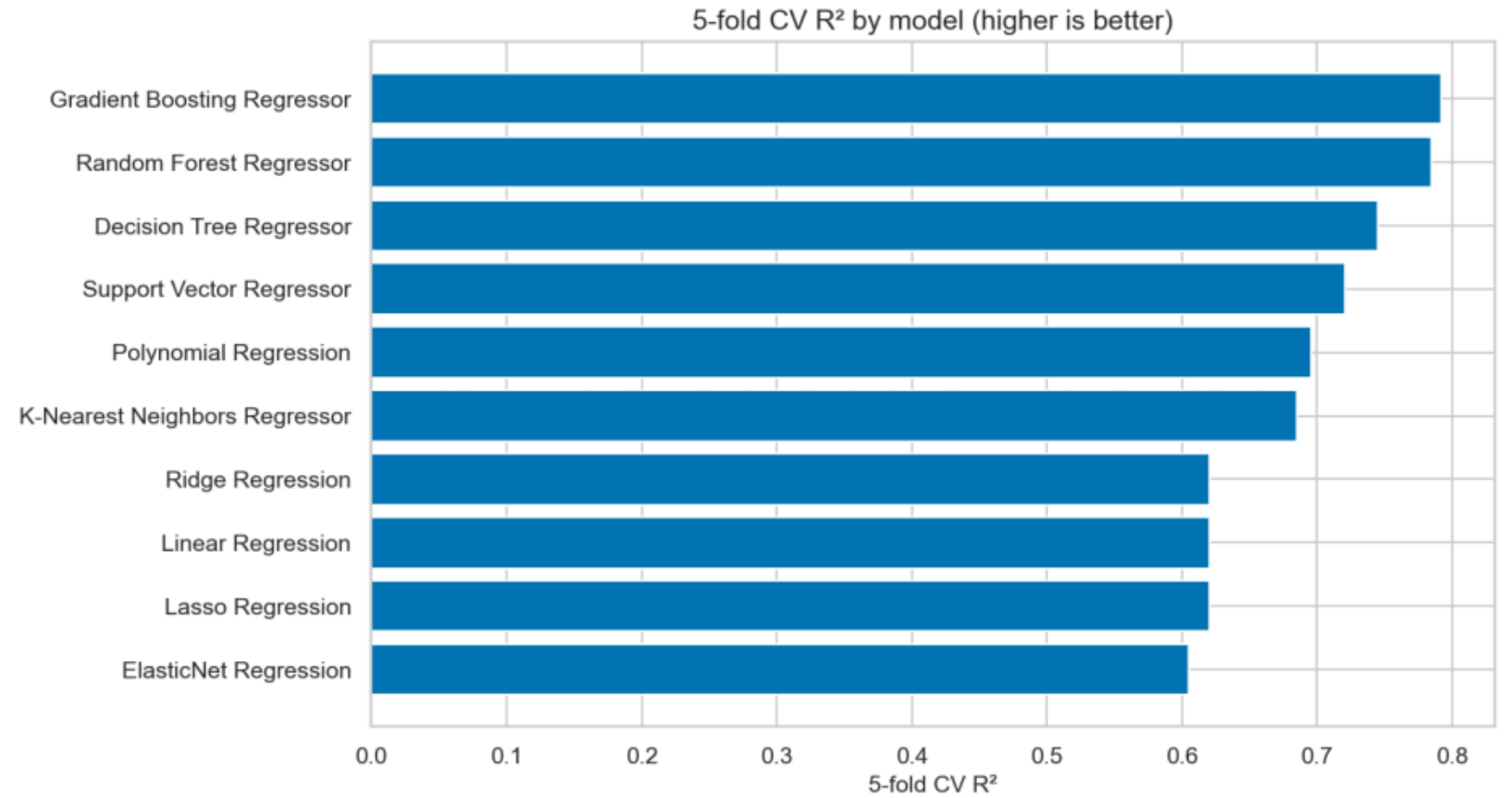
19 test set rmse by model lower is better



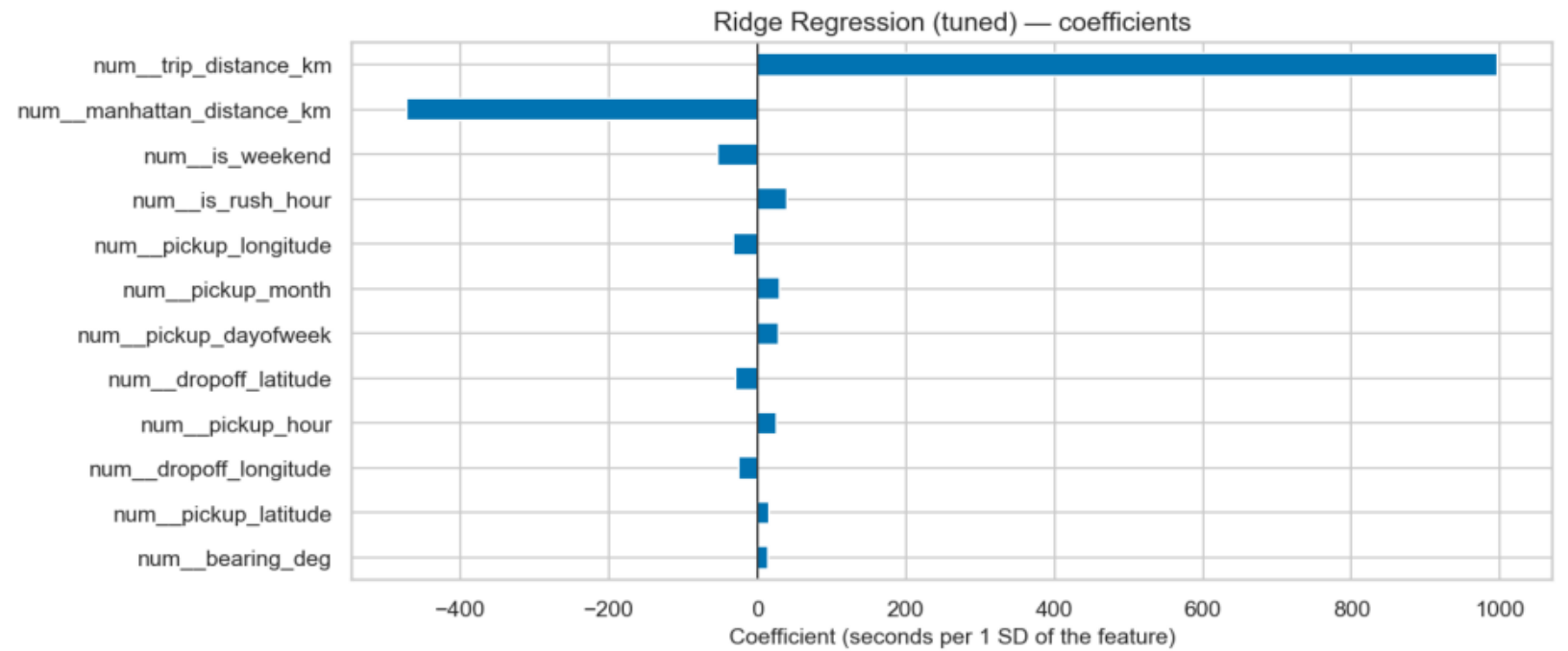
20 test set mae by model lower is better



21 5 fold cv r² by model higher is better



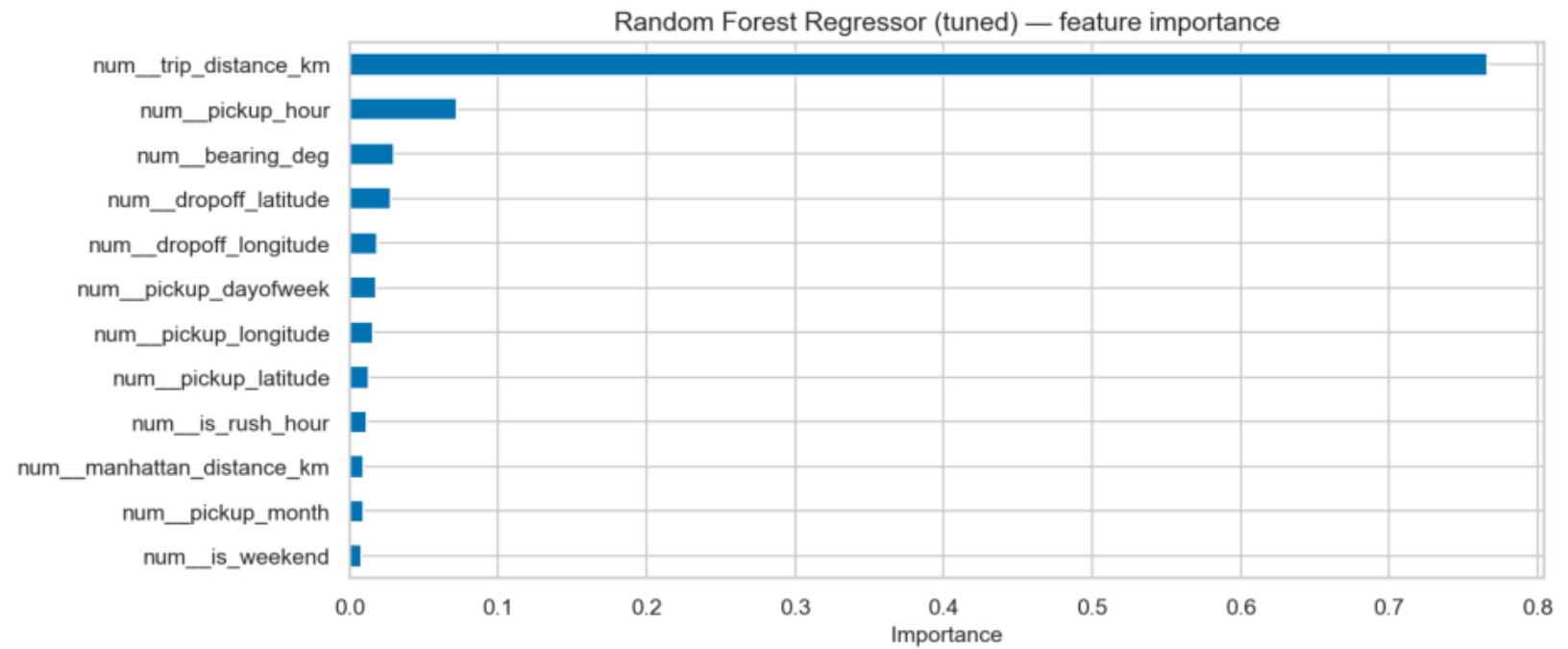
22 ridge regression tuned result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Ridge Regression (tuned)	0.6241	395.8	273.0	0.6208	0.0039

23 random forest regressor tuned result



Mandatory metrics (shared test set)

Model	Test R²	RMSE (s)	MAE (s)	5-fold CV R² mean	5-fold CV R² std
Random Forest Regressor (tu	0.7921	294.3	189.9	0.7860	0.0022

24 predicted vs actual gradient boosting regressor

